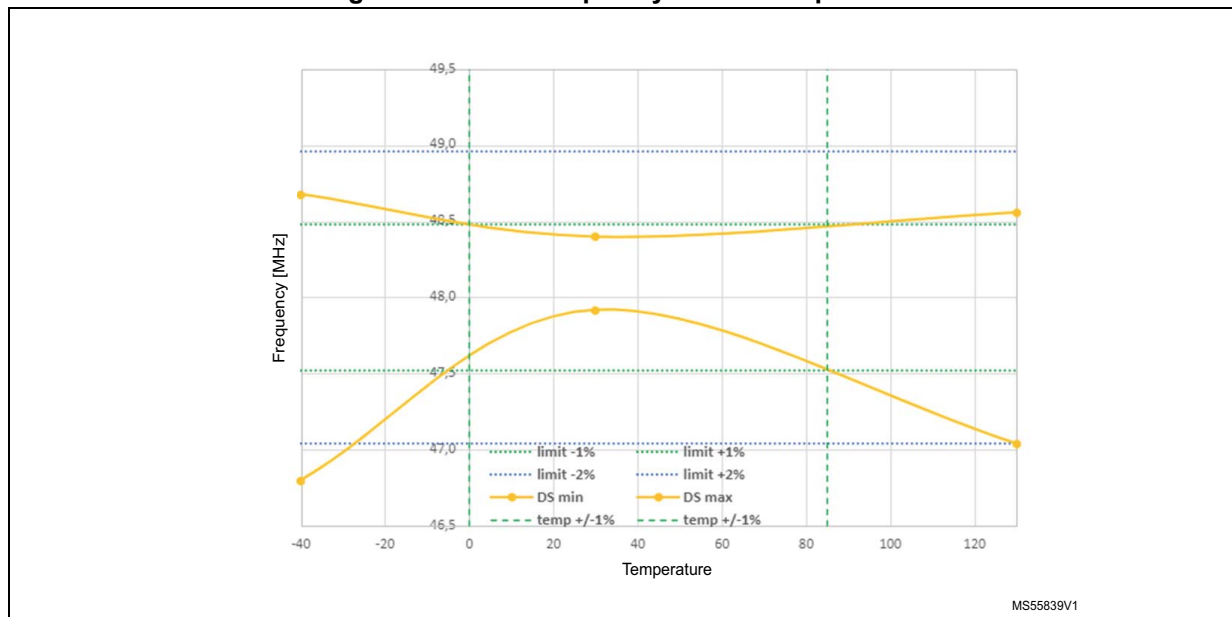


Figure 18. HSI48 frequency versus temperature



Low-speed internal (LSI) RC oscillator

Table 42. LSI oscillator characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{LSI}	LSI frequency	$V_{\text{DD}} = 3.3 \text{ V}$, $T_A = 25 \text{ }^{\circ}\text{C}$	31.04	32	32.96	kHz
		$V_{\text{DD}} = 2 \text{ V to } 3.6 \text{ V}$, $T_A = -40 \text{ to } 125 \text{ }^{\circ}\text{C}$	29.5 ⁽¹⁾	-	34 ⁽¹⁾	
$t_{\text{SU(LSI)}}^{(2)}$	LSI oscillator start-up time	-	-	80	130	μs
$t_{\text{STAB(LSI)}}^{(2)}$	LSI oscillator stabilization time	5% of final frequency	-	125	180	μs
$I_{\text{DD(LSI)}}^{(2)}$	LSI oscillator power consumption	-	-	110	180	nA

1. Evaluated by characterization – Not tested in production.

2. Specified by design – Not tested in production.

5.3.9 flash memory characteristics

Table 43. Flash memory characteristics⁽¹⁾

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
t_{prog}	Word programming time	64 bits	-	85.0	125.0	μs
$t_{\text{prog_row}}$	Row (32 double word) programming time	Normal programming	-	2.7	4.6	ms
		Fast programming	-	1.7	2.8	
$t_{\text{prog_page}}$	Page (2 Kbyte) programming time	Normal programming	-	21.8	36.6	
		Fast programming	-	13.7	22.4	
t_{ERASE}	Page (2 Kbyte) erase time	-	-	22.0	40.0	